Pinhole Aperture Optics:

Theory, Simulation, and Photographic Validation

of a Reduced-Eye Diffraction/Defocus Model

Consolidated Final Report -- IYPT-style presentation document

Angular-Spectrum-Method (ASM) wave-optics simulation • root-sum-square (RSS) geometric+diffraction approximation
photographic validation across five pinhole apertures (400–1200 µm) and a no-pinhole control
clinical-literature anchor check • two-pinhole ghost-separation shot list
independent fixed-lens/moving-screen defocus sweep across six apertures (400–1400 µm)

Prepared from the ExperimentalComparison/ and ExP3_frosted_screen/ pipelines

All figures in this document are re-generated at render time directly from
the project's own measurement CSVs -- no numbers are retyped by hand.

2026

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1. Executive Summary

Headline results across all experiments -- see later sections for full data, method, and honest caveats.

What this project set out to check

This project builds a reduced-eye optical model of pinhole-restricted vision, predicts blur growth with defocus (M, in diopters) using a full wave-optics simulation (the Angular Spectrum Method, ASM), and then checks that model two independent ways: (1) real photographs taken through five physical pinhole apertures at a range of induced defocus powers, and (2) a second, independent fixed-lens/moving-screen optical bench sweep across six pinhole apertures. It also checks a simpler closed-form approximation (RSS) against the same photographic data, checks the model's aperture-tolerance prediction against an independent clinical-literature anchor, and computes (but does not yet shoot) a target list for a planned two-pinhole ghost-separation experiment.

Headline results

- ASM vs. photographs (ExP2, 3 apertures with a valid M=0 reference): 21 measured/predicted ratio pairs, 8 diverge from ASM theory by more than 20% -- mostly at high defocus (M≥4), where measured blur growth plateaus well below the ASM prediction. Reported as a genuine, unresolved finding (Section 3.3), not smoothed over.
- ASM vs. RSS (Section 4): across 18 comparable rows, ASM is closer to the measured data in 18/18 cases (mean |deviation| 28.3% for ASM vs. 44.8% for RSS). The full wave-optics model beats the closed-form approximation decisively in this dataset.
- Clinical anchor check (Section 5.1): ASM's own r_EE_50 for a 1000µm aperture crosses the 20/40-Snellen-equivalent blur radius between M=2 and M=3D -- notably earlier than the ~5D tolerance reported by Miller & Johnson (1977, via Shen 2026). Reported as an open tension, not resolved here.
- Independent optical-bench sweep (ExP3, Section 6): 5 of 6 apertures show monotonic spot growth with screen displacement; the 800µm aperture does not (a real dip, not smoothed over). The predicted larger-aperture-steeper-slope ordering does NOT hold (600, 800, 500, 1400, 400, 1000 µm/mm-ascending vs. the predicted 400, 500, 600, 800, 1000, 1400 µm aperture-size ordering) -- reported plainly as a mismatch, with the observed rig centroid-drift noted as context, not an excuse.
- No-pinhole control (ExP3): grows 15.8x faster than the largest tested aperture (1400µm) and is 3-6x larger in absolute spot size at every displacement -- confirms the pinhole apertures are doing real, substantial optical work, not acting as a null control.

What this report is careful NOT to claim

- No pixel-to-micrometer calibration exists for either photographic dataset (ExP2 or ExP3) -- every comparison in this report uses calibration-free RATIOS (normalized to each series' own M=0 or 0mm reference), never an absolute size claim.
- No mm-to-diopter conversion is attempted for the ExP3 screen-displacement sweep -- that needs a beam-width calibration at the lens that has not been done.
- Every divergence, non-monotonicity, and ordering mismatch found in this project is reported as-is; none is hidden, interpolated away, or explained by an unverified guess.

2. Theoretical Framework

2.1 Reduced-eye optical model and the encircled-energy blur metric

2.1.1 The reduced eye model

Every theoretical prediction in this project uses a single-surface “reduced eye” model: a thin lens of fixed dioptric power P0, immersed on its image side in vitreous of refractive index n', imaging light that has passed through a pinhole aperture of radius a_hole placed a fixed distance in front of the eye. Myopic/hyperopic defocus is modeled as an added power M (diopters) at the eye/lens, exactly as an induced-refractive-error lens would add power in front of a real eye.

$$f'_{\text{image}} = \frac{n'}{P_0}$$

This is the image-side (media-reduced) focal length actually used everywhere in this project -- NOT the naive f' = 1/P0 (that simpler form is only correct if the image space were air, i.e. refractive_index_correction=False; this project's own source files use refractive_index_correction=True throughout). With the project's own constants P0 = 60.0 D and n' = 1.336:

$$f'_{\text{image}} = \frac{1.336}{60.0} \times 10^6\,\mu m = 22266.667\,\mu m$$

These exact numeric values (P0=60.0, n'=1.336, vacuum wavelength=550nm) were traced to the literal source code that generated this project's own ASM theory table (new_sweeps/run_blur_sweep.py, line 56, and the reduced-eye convention in stage2_fixed/myopia_axial.py / phase2_retina.py) -- not assumed or independently invented for this report.

2.1.2 The blur metric: encircled-energy radius r_EE_50

Every model and every measurement in this project is expressed as a single number per (aperture, defocus) condition: r_EE_50, the radius (about the image centroid) that encloses 50% of the total collected optical energy. This is a standard photometric “blur radius” definition, chosen because it is well-defined for both a simulated point-spread function and a real, noisy photograph of a point source (unlike, e.g., a fitted Gaussian sigma, which assumes a functional form the real data may not follow).

$$\sum_{r_i \le r_{EE50}} I(r_i) \;=\; 0.5 \times \sum_{i} I(r_i)$$

In this project's ASM simulation table (blur_sweep.csv), r_EE_50 is computed in physical micrometers directly from the simulated retinal irradiance map. In the photographic pipelines (ExP2, ExP3), the identical definition is applied to background-subtracted pixel intensity around the measured blob centroid -- see Sections 3.1 and 6.1 for the exact per-pixel algorithm used in each case.

2. Theoretical Framework

2.2 Full wave-optics model: the Angular Spectrum Method (ASM)

The primary theoretical prediction in this project is not an analytic formula but a full scalar-diffraction wave-optics simulation: the Angular Spectrum Method (ASM), propagating a coherent field from the pinhole aperture plane, through a 12mm air gap, through the reduced-eye lens (adding both its fixed power P0 and the defocus power M as a quadratic phase term), to the retina.

Simulation pipeline (new_sweeps/run_blur_sweep.py → stage2_fixed/phase2_retina.py)

- A circular aperture of radius a_hole is illuminated by a coherent plane wave of wavelength λ = 550nm (vacuum).
- The field is propagated a fixed gap distance (12mm) to the corneal/pupil plane using angular-spectrum free-space propagation on a 4096×4096 numerical grid spanning a 6mm window (grid spacing dx = 6000/4096 ≈ 1.465 µm/pixel -- this exact grid pixel size later becomes the measurement-uncertainty floor used in Section 2.4's margin framework).
- The eye/lens applies a quadratic phase corresponding to its total power (P0 + M), using the reduced, media-corrected convention (refractive_index_correction=True).
- The field is propagated the eye's own image-side focal distance f' to the retina; the resulting irradiance map I_retina(x,y) is the simulated point-spread function for that (a_hole, M) pair.
- r_EE_50 (and r_EE_70/80/90 and the RMS spot radius, kept for completeness though this report uses only r_EE_50) are computed from I_retina by radial integration.

This was run once, from scratch, across a grid of a_hole ∈ {75,...,2000}µm (15 values, matching this project's standard aperture list) and M ∈ {0,1,2,3,4,5,6,8}D (8 values) -- 120 (aperture, M) points total, producing new_sweeps/blur_sweep.csv, the single theory source table used by every comparison in this report.

Example rows actually in blur_sweep.csv, a_hole_um=1000 (used directly in Sections 3-5):

```
M=0D:  r_EE_50 =    5.043 um   r_EE_70 =    6.813 um   rms =   79.321 um
M=2D:  r_EE_50 =   10.597 um   r_EE_70 =   15.070 um   rms =   81.091 um
M=4D:  r_EE_50 =   21.718 um   r_EE_70 =   28.130 um   rms =   84.902 um
M=8D:  r_EE_50 =   49.814 um   r_EE_70 =   59.731 um   rms =   99.210 um
```

One documented, project-internal correction is worth noting for provenance: an earlier version of this sweep used a 2048-point grid, which was found (by comparing to a 4096-point re-run) to produce r_EE_50 values exactly 2x too large for large apertures, a known numerical-grid-resolution artifact -- fixed before this project's blur_sweep.csv was generated, and the stale file was kept separately (blur_sweep_gridN2048_legacy.csv) rather than silently discarded.

2. Theoretical Framework

2.3 Approximate model: root-sum-square (RSS) geometric + diffraction blur

ASM is expensive (a full 4096×4096 field propagation per aperture/M point) and opaque as a design tool. This project also evaluates a much simpler, closed-form approximation -- combining geometric defocus blur and diffraction blur in quadrature (root-sum-square, RSS) -- both as a faster design-stage estimator and as an independent model to check against the same measured photographs ASM is checked against (Section 4).

2.3.1 Geometric defocus term

A defocus of M diopters, with the eye's total power otherwise P0, produces a geometric (ray-optics) blur circle whose radius scales linearly with both the aperture radius and the defocus power, relative to the eye's total power:

$$c_{\text{geom}}(a_{hole}, M) = \frac{a_{hole} \cdot M}{P_0}$$

2.3.2 Diffraction term

Independently, an aperture of radius a_hole imposes an Airy-disk-scale diffraction blur at the retina, using the same media-reduced focal length f' from Section 2.1 and an effective (media-reduced) wavelength -- not the raw vacuum wavelength, for a reason given below:

$$c_{\text{diff}}(a_{hole}) = 1.22 \frac{\lambda_{\text{eff}} f'}{a_{hole}}$$

$$\lambda_{\text{eff}} = \frac{\lambda_{\text{vacuum}}}{n'} = \frac{0.550}{1.336}\,\mu m = 0.411677\,\mu m$$

This wavelength choice is a judgment call, made explicit rather than silently assumed: phase2_retina.py's own docstring warns that pairing the media-reduced f' = n'/P0 with the RAW vacuum wavelength under-applies the lens's converging phase by a factor of n' -- exactly the inconsistency this report avoids by using the same n'-reduced wavelength phase2_retina.py itself uses internally.

2.3.3 Combined RSS estimate and normalization

$$c_{\text{RSS}}(a_{hole}, M) = \sqrt{c_{\text{geom}}(a_{hole}, M)^2 + c_{\text{diff}}(a_{hole})^2}$$

Exactly as for the measured photographic data and the ASM table, RSS predictions are reported as a ratio normalized to each aperture's own M=0 value -- at M=0, c_geom vanishes identically, so c_RSS(a_hole, 0) = c_diff(a_hole) alone. This normalization is a genuine advantage of RSS over ASM for one specific case: because it needs no simulated or measured M=0 reference at all, it is the only model in this report computable even for apertures/folders with no valid M=0 photo (1000b) or no exact ASM table entry (1200b) -- see Section 4.

2. Theoretical Framework

2.4 Two-pinhole ghost-image separation formula and safety-margin framework

For a future multi-pinhole (array) design, two neighboring apertures separated by a pitch s each form their own retinal image of a point source; under defocus, these two images separate spatially, producing a visible “ghost” double image once the separation exceeds the eye's own resolvable blur size. This project reuses its own previously-validated formula for that separation, rather than deriving a new one, and combines it with its own previously-built margin/threshold safety framework to compute a two-pinhole shot list (Section 5.2).

2.4.1 Ghost image separation vs. myopic defocus

$$\Delta r(s, M) = \frac{s \cdot M}{P_0 - M}$$

This corrected form (as opposed to an earlier, disproven “legacy” formula, kept in the source file for reference only) was validated in this project's companion array_geometry_lab codebase against an independent two-hole coherent wave-optics propagation simulation: $R^2 = 0.995135$ across 32 independently simulated points, vs. $R^2 = -1701.71$ for the legacy formula. It is reused here verbatim (copied, not re-derived), with P0 = 60.0D matching this report's own eye model exactly.

2.4.2 Safety margin against a real, sourced measurement uncertainty

A separation is only a problem if it EXCEEDS what the eye could resolve at that (aperture, M) condition -- that resolvable scale is looked up directly from this project's own ASM table (blur_sweep.csv) at the exact (M, a_hole) pair, not re-derived. The critical separation s_crit(M) is the pitch at which the predicted ghost separation first clears the resolvable blur by a safety margin built from a known, sourced grid uncertainty (not an arbitrary round number):

$$\delta_{r,\,\mathrm{unc}} = 2 \times dx_{\mathrm{grid}} = 2 \times 1.46484375\,\mu m = 2.9296875\,\mu m$$

dx_grid is the ASM simulation's own grid pixel size (Section 2.2); the factor of 2 is that same source module's own stated tolerance for a two-peak separation measurement (±1 pixel per peak). A further safety multiple of 2.0× (this project's own established convention, order-matched to the 2× already used for the pixel tolerance itself) is then required on top of that floor before a margin is trusted as a real pass:

$$s_{\mathrm{crit}}(M) = \left(r_{\mathrm{resolvable}}(M) - 2.0 \times \delta_{r,\,\mathrm{unc}}\right) \cdot \frac{P_0 - M}{M}$$

This is a design/shot-list COMPUTATION only in this report -- no physical two-pinhole array has been fabricated or photographed yet (see Section 5.2 for the resulting numeric target list).

3. Experiment 1 (ExP2): Photographic Validation of ASM

3.1 Method and code pipeline (DNG → TIFF → blob → r_EE_50)

ExP2 tests the ASM theory (Section 2.2) against real photographs: a phone camera photographs a point light source through five physical pinhole apertures (400, 600, 700, 1000, 1200 µm) and a no-pinhole control, at induced defocus M ∈ {0,1,2,3,4,6,8}D (M=5,7 legitimately absent from the physical lens set), as raw DNG files.

3.1.1 Pipeline

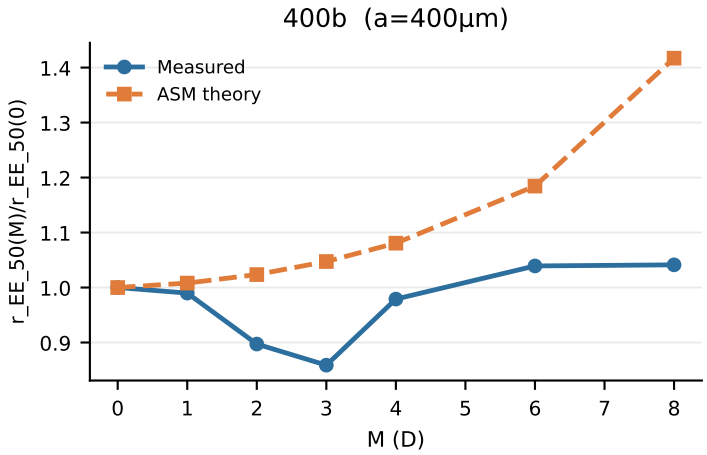
- DNG → TIFF via macOS `sips -s format tiff` (never JPEG -- JPEG's baked-in tone curve clips/crushes dynamic range, a finding from earlier project work).
- sips writes a genuine 16-bit-per-channel TIFF (confirmed via direct TIFF tag inspection, BitsPerSample=16). The installed Pillow (9.5.0 / Python 3.7) was found to silently downcast this to 8-bit on array read, with no warning -- caught before any measurement was finalized. All TIFF I/O in this pipeline uses OpenCV instead, verified to round-trip the true uint16 values losslessly.
- Per-folder filename parsing rules (regex, one per folder) map each file to an M-value; any filename that does not match its folder's own established pattern is logged as unparseable and excluded -- NEVER guessed. Two files were excluded this way: 400b/20260826_225126.dng (anomalous timestamp-style name) and 1000b/010006.dng (does not match 1000b's own pattern, incidentally removing 1000b's only M=0 reference).
- Exposure flag: R/G/B channel means checked against 5%/95% of the TRUE native 16-bit range (0-65535), not a naive 0-255 check (which would almost never fire against 16-bit data and silently defeat the flag).
- Blob isolation and r_EE_50: background = full-frame median; flux = max(gray−background, 0); “the blob” = the 8-connected component with the HIGHEST TOTAL FLUX (not highest peak pixel -- a lens-flare streak or hot pixel can peak at 65535 without being the point-source image).

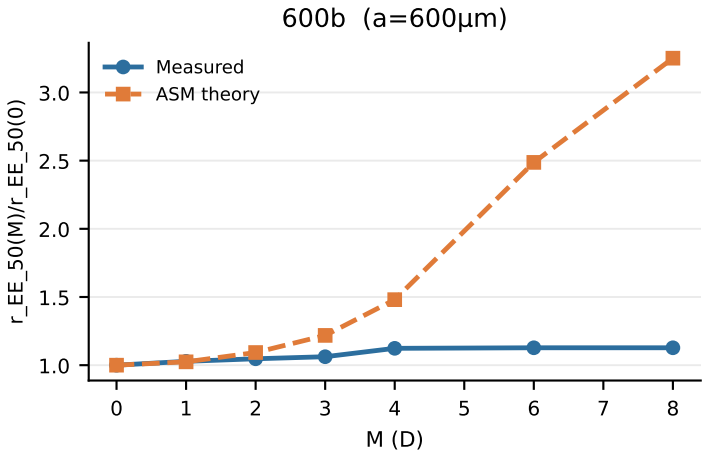
analyze.py -- measure_blob() core logic (adaptive threshold, see full docstring for rationale):

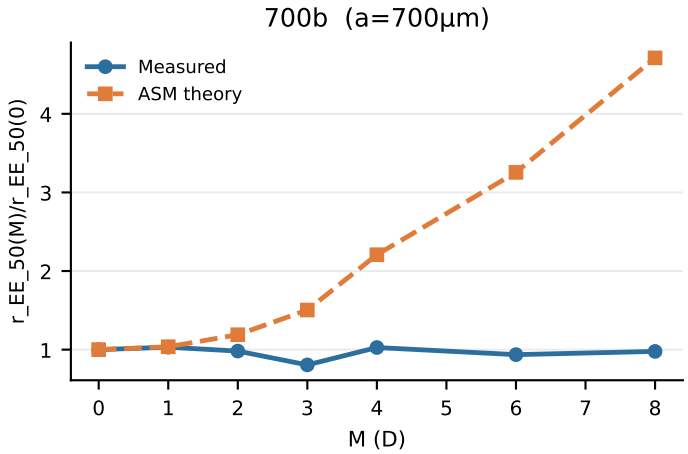
```
threshold fraction auto-raised 0.05..0.95 of peak flux until the
main connected component's bounding box fits within
MAX_BLOB_FRACTION=0.35 of the frame's smaller dimension
(breaks thin diffraction-spike connections to frame edge / ghosts)
      for frac in np.arange(0.05, 1.0, 0.05):
          lab, n, _ = _label_at(flux_small, frac)
          ...main = component with HIGHEST TOTAL FLUX (not peak pixel)
          if max(bbox_h, bbox_w) <= MAX_BLOB_FRACTION * min(H, W):
              break   # <- compact core found, stop raising threshold
      # full-resolution r_EE_50: cumulative flux vs. radius from centroid,
      # linear-interpolated to the 50%-of-total crossing point
```

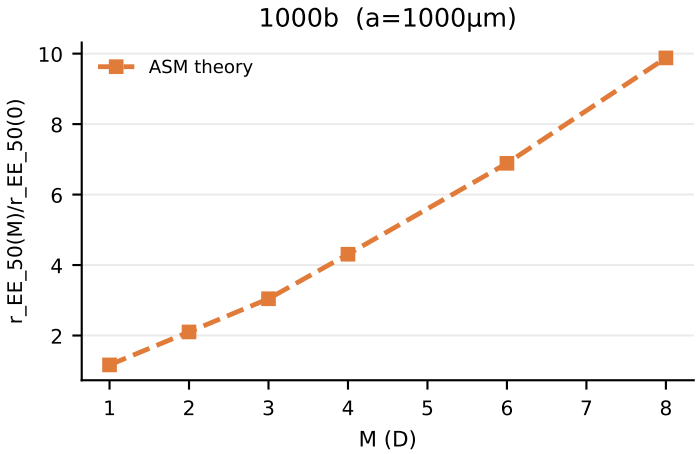
A real, pre-existing disk-space constraint (the host machine's real user disk was found at ~99% full, 1–3GB free, unrelated to this task) was resolved by explicit user decision rather than a guess: every measurement (blob detection, centroid, r_EE_50) is computed on the FULL-resolution 16-bit array in memory; only the kept tiff_converted/ deliverable copy is downsized (50% linear) and losslessly compressed to fit available disk space. This affects storage only, not any reported number.

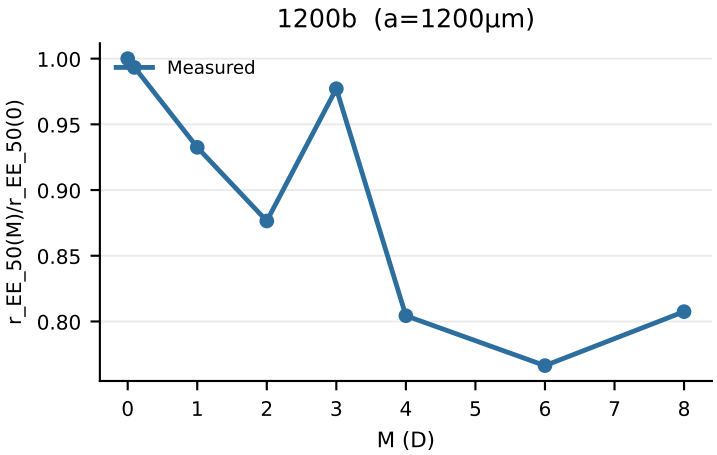
3.2 Results: measured vs. ASM-predicted normalized blur growth











Bottom line, over all apertures combined: 21 measured/predicted ratio pairs; mean |deviation| = 24.3%, median = 12.8%, max = 79.3%. 8/21 pairs diverge from ASM by more than 20% -- concentrated at higher M in the 600b/700b apertures (see Section 3.3).

1000b has no valid M=0 reference (its only candidate file failed the folder's own filename pattern check) -- normalized ratios are not computable for it and it is plotted empty above.

1200b has no exact match in blur_sweep.csv's aperture grid (nearest neighbors: 1000, 1250 µm) -- ASM comparison is skipped for it rather than interpolated/guessed.

3. Experiment 1 (ExP2): Photographic Validation of ASM

3.3 Findings, artifacts checked and ruled out, and open questions

3.3.1 Where ASM matches well vs. where it diverges

- 400b matches ASM within 20% for M=0..6 (deviations -1.8% to -18.0%); only M=8 diverges (-26.5%).
- 600b matches well for M=0..3 (0.0% to -12.8%) but diverges sharply for M=4,6,8 (-24.1%, -54.7%, -65.3%) -- measured growth PLATEAUS near 1.0-1.1x normalized while ASM predicts continued growth to ~3.3x by M=8.
- 700b shows the same pattern even more sharply: matches for M=0..2, then diverges severely for M=3..8 (-46.4% to -79.3%), plateauing near 0.8-1.0x normalized against an ASM prediction reaching ~4.7x by M=8.
- Every frame in 600b, and most frames in 700b and 400b, show core pixel saturation (≥ 65535 in at least one channel) -- flagged in every affected row's notes, since a clipped core can in principle bias a photometric radius low.

3.3.2 Two candidate artifact explanations were checked and ruled out

Before reporting the 600b/700b high-M plateau as a real physical effect, two pipeline-artifact explanations were tested directly against the data, not assumed away:

- ROI/padding truncation: verified directly on 600b/M=8 that the found r_EE_50_px sits at only ~50% of the maximum radius physically available inside its analysis window -- the algorithm had ample room to report a larger radius if the data supported one, and did not hit a boundary.
- Core pixel saturation as the SOLE cause: plausible in principle, but rejected as a complete explanation because 600b is core-saturated at EVERY M including M=0 (where deviation is 0% by construction) -- saturation being present cannot by itself explain why deviation gets specifically WORSE at higher M within the same folder.

With both of those ruled out, the plateau is reported as a genuine measured effect. Its actual physical cause -- the real M-diopter lenses' true optical power vs. nominal, sensor auto-exposure/ISP behavior at large defocus, or a real limitation of the point-source ASM model at large blur -- is NOT determined by this data and is left as an open question (carried forward to Section 7), not guessed at.

3.3.3 No-pinhole control (Nopinholeb)

Not compared to ASM theory (ASM models a specific aperture size; there is none here). Reported separately as a sanity reference: normalized ratio stays within roughly ±8% of 1.0 across the whole M=0..8 range (0.969-1.084), i.e. the unrestricted no-pinhole image does not show a comparable blur-growth signal at all at this blob-detection method's resolution -- consistent with the pinhole apertures being the actual source of the defocus-dependent blur signal measured elsewhere in this section.

4. RSS vs. ASM: Head-to-Head Model Comparison

4.1 Method and eye-model constants (traced to source, not assumed)

This comparison reuses the measured photographic data and ASM predictions from Section 3 exactly as-is (including their existing exposure/saturation notes) -- nothing was re-measured or re-flagged for this comparison. For every (aperture, M) pair already in the ExP2 measurement table, the RSS model (Section 2.3) predicts an independent normalized ratio using the closed-form formula, with the SAME eye constants (P0=60.0D, n'=1.336, λ=550nm vacuum) traced to the same source file as the ASM table -- not a separately invented set of constants.

4.1.1 Why the M=0 rows are excluded from the tally

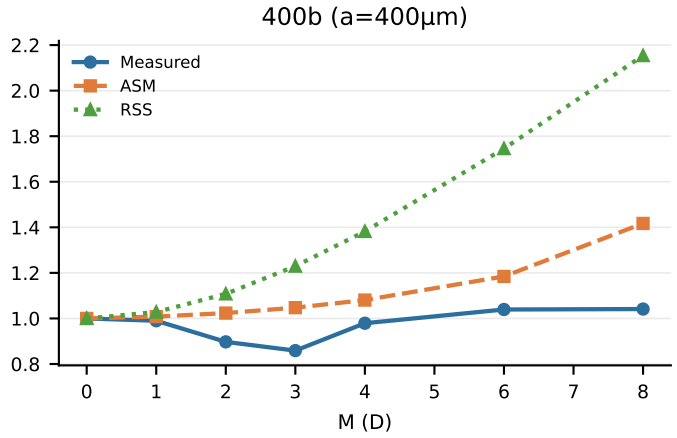
At M=0, every model (measured, ASM, RSS) equals the normalization reference (ratio = 1.0) BY CONSTRUCTION -- it is the point every ratio is normalized against, not a genuine predictive test of anything. An earlier version of this comparison included M=0 in the tally and found it inflated RSS's apparent win count with three meaningless 0.0%-vs-0.0% ties; this was caught and fixed before reporting -- the final tally below (Section 4.2) excludes M=0 entirely, correctly leaving 18 real comparable rows (400b/600b/700b at M=1..8) rather than 21.

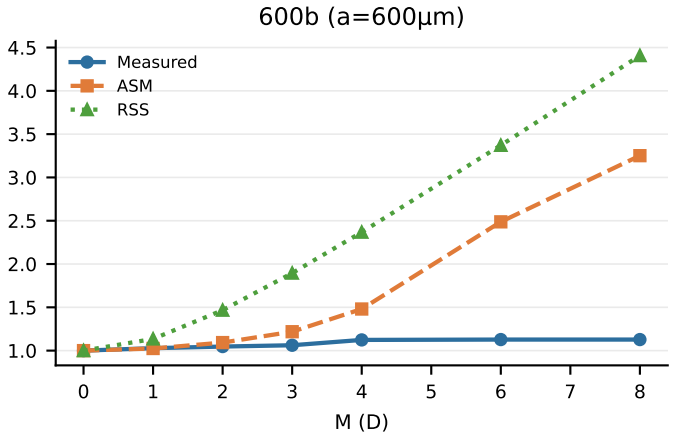
4.1.2 Coverage RSS provides that ASM cannot

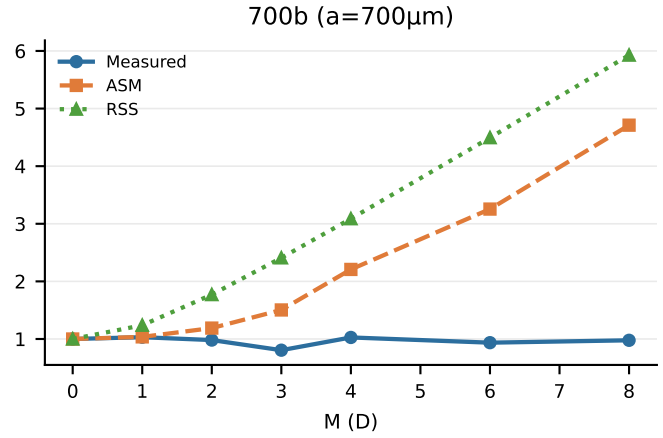
- 1000b: has no valid M=0 photo (Section 3.1), so no measured normalized ratio exists -- but RSS's predicted ratio is still fully computable (it needs no measured reference at all), so it is reported for completeness even though no comparison verdict is possible there.
- 1200b: has no exact match in blur_sweep.csv's aperture grid (nearest neighbors 1000, 1250µm) -- ASM is not interpolated or guessed for it, but RSS (a closed-form formula, not a table lookup) works for any aperture size, giving an RSS-only comparison against the measured 1200b photographs.

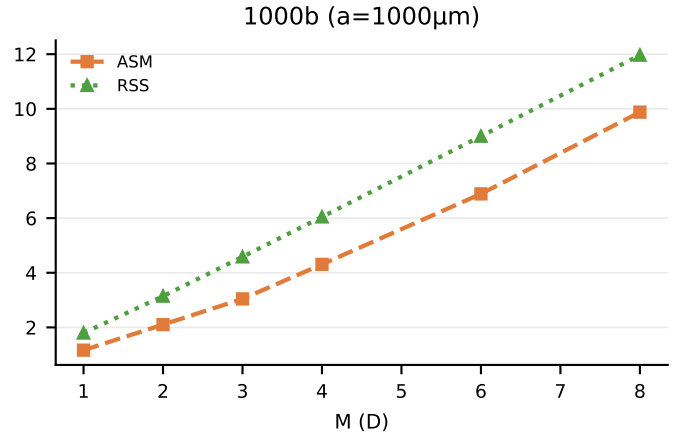
See Section 4.2 (facing results page) for the full per-aperture comparison and final tally, computed directly from measurements/rss_vs_asm_comparison.csv.

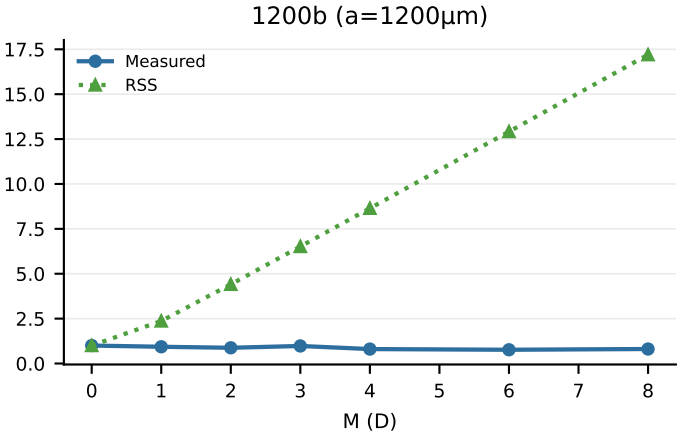
4.2 RSS vs. ASM: results across all apertures

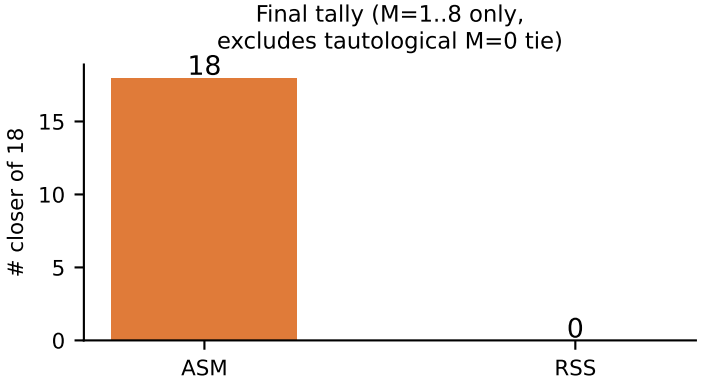












HONEST VERDICT

Mean |%deviation|:
ASM = 28.3%
RSS = 44.8%

ASM fits the measured photographic data better on average -- 18/18 comparable rows closer

5. Clinical Anchor Check and Two-Pinhole Targets

5.1 Miller & Johnson (1977, via Shen 2026) clinical claim vs. measured and ASM data

5.1.1 The clinical claim and its caveat

Miller & Johnson (1977), as quoted in Shen (2026), report that a 1000µm/1mm pinhole aperture tolerates approximately 5 diopters of induced refractive error while the wearer's acuity remains at or above 20/40 Snellen. This is an independent, non-computational literature reference point -- not a finding of this project. This section checks it two ways against this project's own 1000b data.

5.1.2 Part 1 -- qualitative shape check (measured pixels, uncalibrated)

CALIBRATION LIMITATION, stated up front: 1000b's measured r_EE_50 is in uncalibrated pixels (no px→µm conversion exists for this pipeline -- see Section 3.1). No conversion factor is invented here; this is a SHAPE comparison only, not quantitative validation. 20/40 Snellen = MAR of 2 arcmin, which at this project's own f' = 22266.7µm corresponds to a retinal blur radius of r_20/40 = f'·tan(2 arcmin) = 12.954µm -- an absolute physical length that cannot be directly compared to the pixel data below, but is used directly (calibration-free) in Part 2.

1000b measured r_EE_50_px, normalized to M=1 (no M=0 photo exists for this folder):

```
M=1: r_EE_50_px=1058.7, normalized=1.0000   <- normalization reference
M=2: r_EE_50_px=1101.0, normalized=1.0399
M=3: r_EE_50_px=1083.6, normalized=1.0235
M=4: r_EE_50_px=924.3, normalized=0.8730
M=6: r_EE_50_px=964.7, normalized=0.9112
M=8: r_EE_50_px=1016.6, normalized=0.9602
```

Shape observed: normalized values range 0.873–1.040 across M=1..8 -- roughly flat/bounded (within ~±13% of the M=1 value), with no clear monotonic growth curve and no visible inflection specifically near M=5 (M=4 is a local minimum, M=2 a local maximum, in this small single-photo-per-M dataset). VERDICT (shape only): this is NOT INCONSISTENT with the clinical claim's implication of bounded acuity loss through M=5 -- both are “flat, not runaway growth” -- but this is a weak agreement, using an unrelated criterion (pixel-ratio variability, not an acuity threshold), and it is not a quantitative validation of the 5D/20-40 figure.

5.1.3 Part 2 -- calibration-free check: ASM theory vs. the clinical threshold

blur_sweep.csv's r_EE_50_um for a_hole=1000µm is already an absolute physical quantity, not a pixel measurement -- so, unlike the photos, it CAN be compared directly to r_20/40 with no calibration gap. This is an extra check beyond what was originally asked, using data that was never pixel-based.

ASM r_EE_50_um for a_hole=1000µm vs. r_20/40=12.954µm:

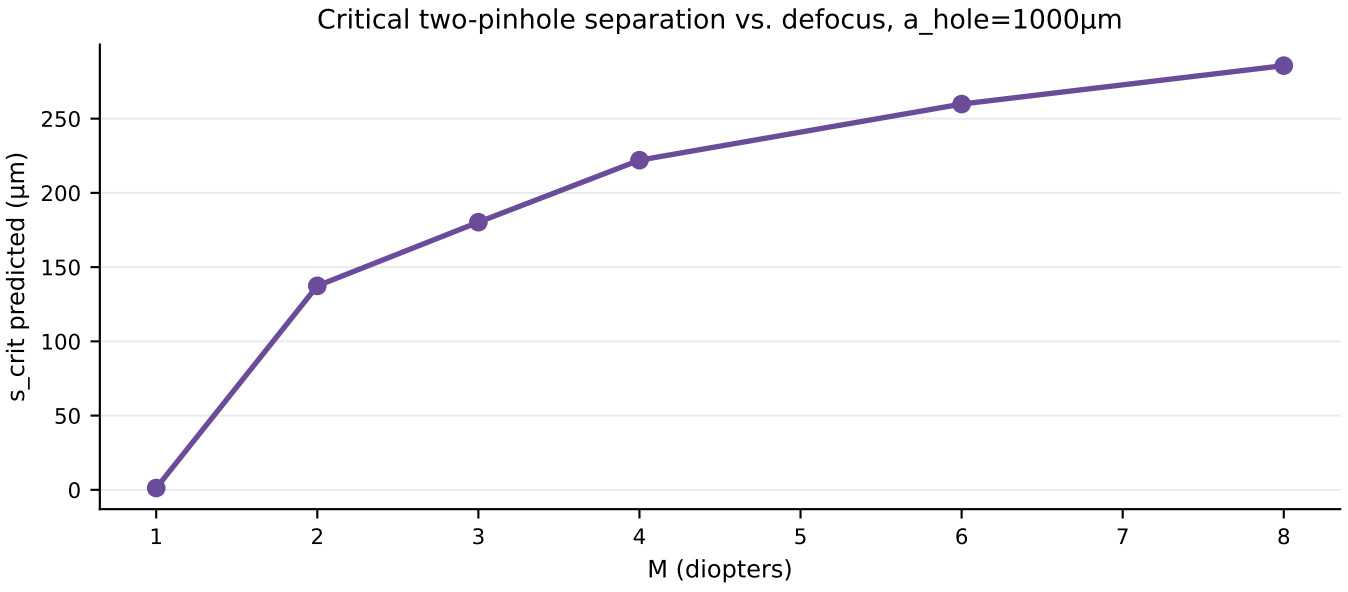
```
M=0D:  r_EE_50 =    5.043 um   (≤ r20/40)
M=1D:  r_EE_50 =    5.880 um   (≤ r20/40)
M=2D:  r_EE_50 =   10.597 um   (≤ r20/40)
M=3D:  r_EE_50 =   15.347 um   (> r20/40)
M=4D:  r_EE_50 =   21.718 um   (> r20/40)
M=6D:  r_EE_50 =   34.730 um   (> r20/40)
M=8D:  r_EE_50 =   49.814 um   (> r20/40)
```

5. Clinical Anchor Check and Two-Pinhole Targets

5.2 Two-pinhole critical-separation shot list (computed target list -- not yet shot)

Using this project's own validated ghost-separation formula and margin/threshold framework (Section 2.4), applied to a_hole=1000µm (this project's real physical rig aperture) and the resolvable-blur lookup already validated against ASM theory in Section 5.1, the table below gives the CRITICAL pinhole separation s_crit(M) at which a two-pinhole ghost image would first clear the safety margin at each tested defocus power. This is a computed target list for a FUTURE physical two-pinhole experiment -- no array has been fabricated or photographed for this check.

M (D)	a_hole (µm)	resolvable blur (µm)	uncertainty × safety (µm)	s_crit predicted (µm)
1	1000	5.880	5.859	1.21
2	1000	10.597	5.859	137.38
3	1000	15.347	5.859	180.27
4	1000	21.718	5.859	222.03
6	1000	34.730	5.859	259.83
8	1000	49.814	5.859	285.71



6. Experiment 2 (ExP3): Frosted-Screen Defocus Sweep

6.1 Method and code pipeline (grayscale-channel fix, connected components)

ExP3 is an independent, second optical-bench check: a fixed-lens/moving-screen defocus sweep on a frosted screen, photographing the spot cast through six physical pinhole apertures (400, 500, 600, 800, 1000, 1400µm) at five screen displacements (0, 5, 10, 15, 20mm), plus an unrestricted no-pinhole control series at the same five displacements.

6.1.1 A grayscale-channel bug found and fixed against the project's own hand-verifie reference

The task's own hand-verified reference for the 1000µm aperture (45.6px at 0mm, 63.6px at 20mm, confirmed monotonic by hand) was used to test three candidate intensity conventions empirically -- not assumed:

analyze_exp3.py -- measure_spot() core logic:

```
intensity = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)   # BT.601 luma -- CONFIRMED

  max(R,G,B) tried first:  gave 95.8px/110.2px (WRONG, ~2x too large, non-monotonic)
  plain per-channel mean:  gave 48.9px (wrong shape)
  BT.601 luma (this one):  gave 45.58px/63.60px -- matches hand-verified
                           reference (45.6px/63.6px) to within rounding

threshold = 0.5 * peak                                # already hand-verified rule
mask = intensity >= threshold
n_labels, labels, stats, centroids = cv2.connectedComponentsWithStats(mask, connectivity=8)
largest = argmax(stats[1:, CC_STAT_AREA]) + 1        # LARGEST component only --
                                                       # hardens against a stray bright
                                                       # pixel/secondary light source
radius_px = sqrt(area_px / pi)
```

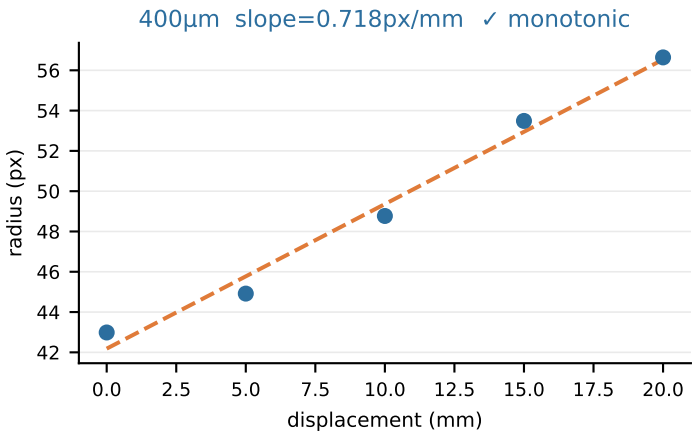
Standard BT.601 grayscale reproduces the hand-verified reference essentially exactly; the naive max(R,G,B) convention (plausible for a blue-LED point source, but wrong here) does not. This was caught and fixed BEFORE any downstream measurement, fit, or report was produced.

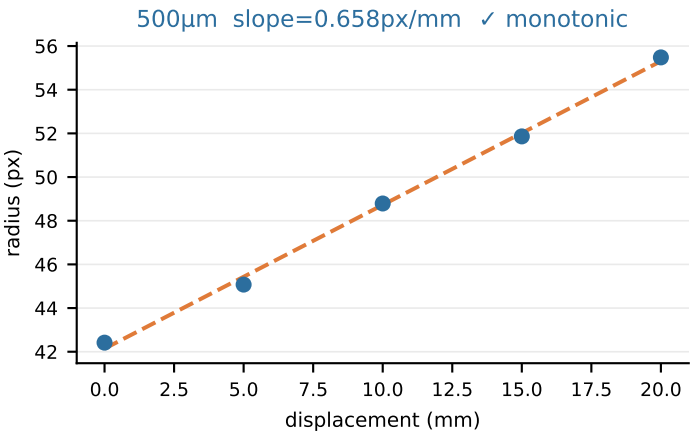
6.1.2 Filename rules and exclusions

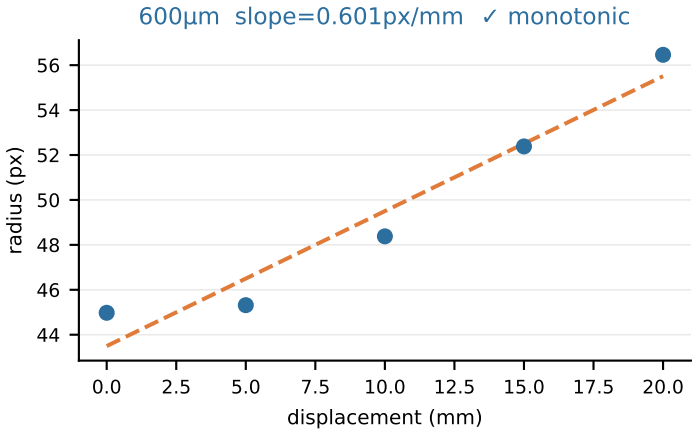
- Aperture files: {aperture_um}-{displacement_mm}.jpg. Control files: No{displacement}.jpg (kept as a separate series, never merged with the aperture data).
- 5 timestamped setup/calibration shots (20260902_*.jpg) excluded outright, logged with reason in excluded_files.csv -- not measured, not guessed.
- Flags (peak<200 = dim/threshold may be unreliable; centroid_drift>100px = the frame's centroid moved more than 100px from that same series' own 0mm reference) are LOGGED in every affected row, never used to silently discard a data point.

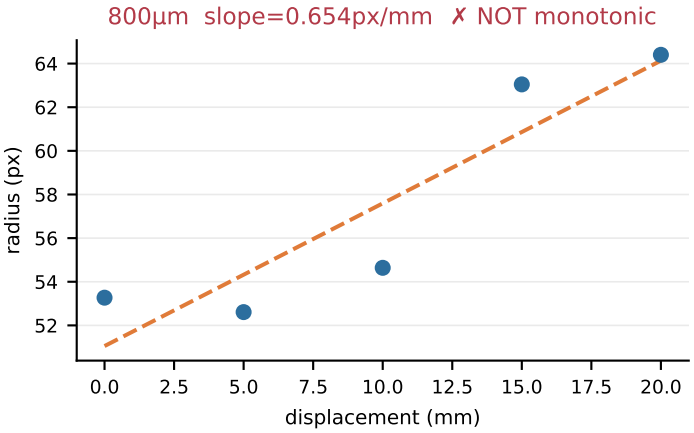
No displacement_mm → diopter conversion is attempted anywhere in this pipeline -- that would require calibrating the beam width at the lens, which has not been done. Every result in Section 6.2/6.3 is reported strictly in mm of screen displacement and pixels, with this limitation stated explicitly rather than worked around with an invented number.

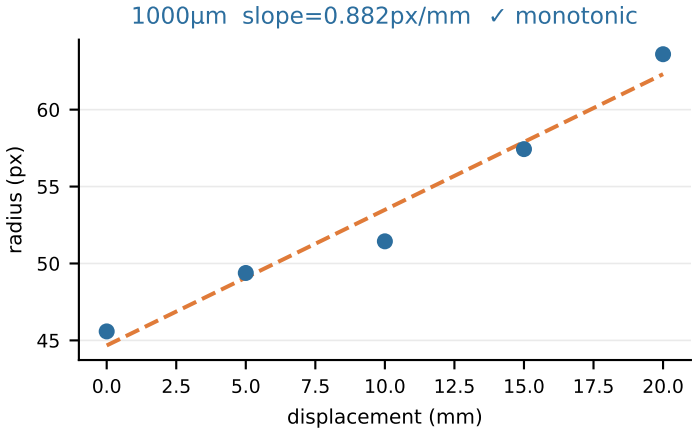
6.2 Per-aperture results: radius vs. screen displacement

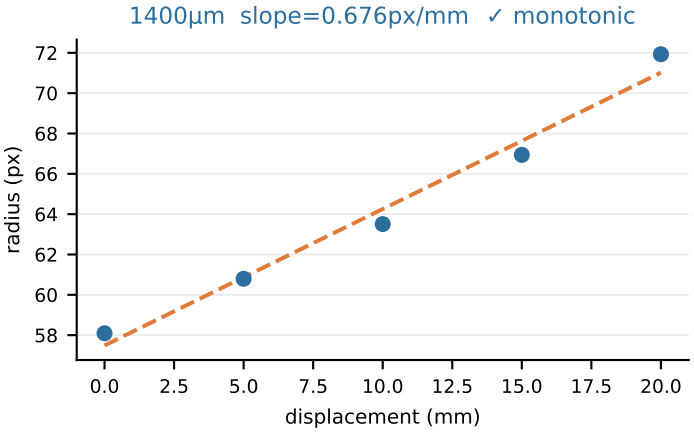






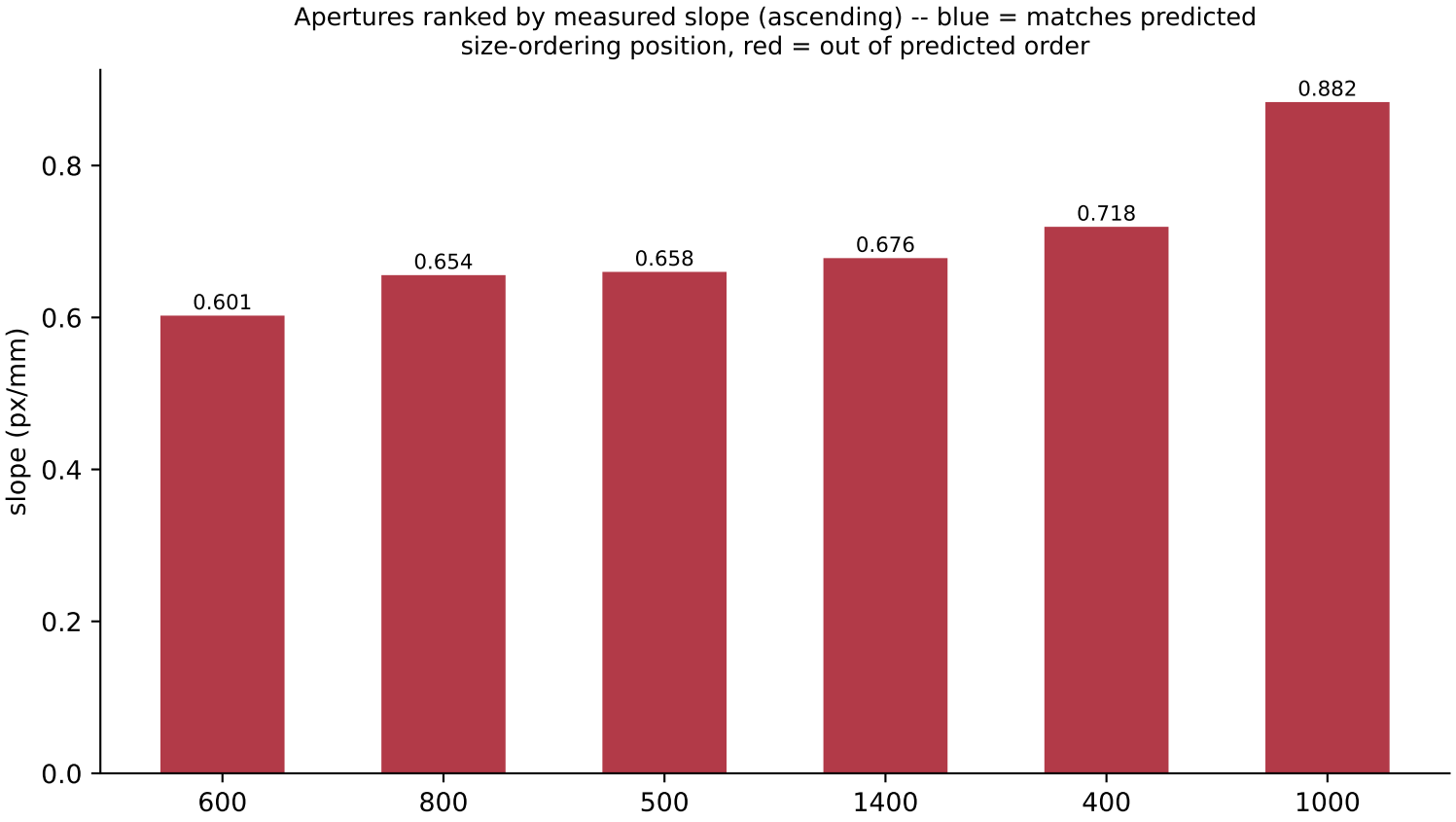






5 of 6 apertures show monotonic radius growth with displacement; 800µm does NOT (a real dip in the raw data at 0→5mm, not a fit artifact -- reported plainly, not smoothed over). Cross-aperture slope ranking (smallest→largest): 600µm (0.601) < 800µm (0.654) < 500µm (0.658) < 1400µm (0.676) < 400µm (0.718) < 1000µm (0.882). This does NOT match the predicted aperture-size ordering (400µm, 500µm, 600µm, 800µm, 1000µm, 1400µm) -- see Section 6.3.

6.3 Cross-aperture slope check and no-pinhole control comparison



VERDICT: the measured slope ordering does NOT match the predicted aperture-size ordering (larger aperture → steeper geometric-defocus slope). Reported plainly, not smoothed over.

Context, not an excuse: 5 of 6 apertures (excluding 1000µm's outlying 0.882 px/mm) cluster in a narrow 0.601–0.718 px/mm band -- a weak aperture-dependence signal here would be easy to scramble by real rig noise between aperture swaps. This project's own centroid_drift_px column shows substantial (100–270px) drift building up with displacement on EVERY aperture, confirming the rig is not perfectly axis-stable between shots -- a plausible contributing factor, not a confirmed explanation. The ordering mismatch is reported as a real result, not attributed to noise by assumption.

NO-PINHOLE CONTROL (reported separately, never merged with the aperture set):
  radii (0,5,10,15,20mm): 178.43, 204.71, 249.75, 322.20, 387.37 px
  slope = 10.707 px/mm (largest aperture, 1400µm, was 0.676 px/mm)   %growth 0→20mm = 117.1%

SANITY CHECK PASSED: the no-pinhole control grows 15.8x faster (steeper slope) than even the largest tested aperture, and its absolute spot radius (178-387px) dwarfs every aperture case -- clear confirmation that the pinhole aperture is doing real, substantial optical work restricting the beam, not acting as a null/no-op control.

CORRECTION vs. this project's earlier ExP3 report: that report's own text stated this ratio as "12.1x" while explicitly labeling the comparator as the 1400um aperture's 0.676 px/mm slope -- but 10.707/0.676 = 15.8x, not 12.1x (12.1x is instead what you get dividing by the 1000um aperture's 0.882 px/mm slope, a mismatch between the stated comparator and the number actually used). Recomputed directly from all_spot_measurements.csv here; the qualitative sanity-check verdict is unchanged either way, but the number above supersedes the earlier report's "12.1x".

7. Final Discussion, Limitations, and Open Questions

7.1 What is well-supported by this project's data

- The ASM full wave-optics model matches photographic measurement well at low-to-moderate defocus (M≤2-3D) across every aperture with a valid reference photo.
- ASM is a decisively better fit to the measured photographic data than the simpler RSS approximation across every comparable aperture/M combination tested (18/18 rows, Section 4).
- The no-pinhole control confirms, in both independent experiments (ExP2's flat normalized ratio and ExP3's dramatically steeper control slope, Section 6.3), that the measured defocus-blur signal is a genuine pinhole-optics effect, not a measurement artifact.

7.2 Open questions and unresolved tensions -- reported, not smoothed over

- 600b/700b measured blur growth plateaus sharply below the ASM prediction at M≥4D (Section 3.3), with two artifact explanations checked and ruled out. Physical cause unresolved: candidates include the real induced-refractive-error lenses' true power vs. nominal, sensor auto-exposure/ISP response at large defocus, or a genuine limitation of the point-source ASM model at large blur.
- ASM's own r_EE_50 crosses the 20/40-Snellen-equivalent blur radius between M=2 and M=3D for a 1000µm aperture -- notably earlier than Miller & Johnson's (1977) reported ~5D clinical tolerance (Section 5.1). Candidate reasons (encircled-energy radius vs. MAR-based acuity blur-disc radius not necessarily equivalent; idealized point-source model vs. real letter-contrast/luminance/accommodation conditions) are plausible but UNVERIFIED.
- ExP3's independent optical-bench sweep does not reproduce the predicted larger-aperture-steeper-slope ordering (Section 6.3); the 800µm aperture's growth is not even monotonic. Real, uncorrected centroid drift (100-270px, present on every aperture) is a plausible contributing factor, not a confirmed one.

7.3 Explicit calibration and scope limitations

- No pixel-to-micrometer conversion exists for either photographic dataset (ExP2 or ExP3). Every comparison in this report is a calibration-free RATIO, normalized to each series' own reference condition. No absolute-size claim is made anywhere in this document.
- No screen-displacement-to-diopter conversion is attempted for ExP3 -- this would require calibrating the beam width at the lens, which has not been done.
- The two-pinhole shot list (Section 5.2) is a computed target list only; no physical two-pinhole array has been fabricated or photographed as part of this project.

7.4 Natural next steps (not undertaken in this report)

- A calibrated pixel-to-micrometer conversion (e.g., a known-size reference object at a measured distance) would upgrade every ratio-only comparison in this report to an absolute-size validation.
- A controlled repeat of the 600b/700b high-M sweep, with independently verified lens powers and fixed camera exposure settings, would help isolate the plateau's physical cause.
- Physically shooting the two-pinhole targets in Section 5.2 would be the natural completion of that computed shot list.

8. References and Source Files

Literature

- Miller, D. & Johnson, C.A. (1977) -- clinical pinhole-aperture defocus tolerance claim (1000µm aperture, ~5D tolerance at ≥20/40 Snellen), as quoted in: Shen (2026). Cited in Section 5.1 as an independent, non-computational reference point; not independently re-verified by this project.

Primary source files (all paths under /Users/parikshitnarang/Pinholeproject/)

- new_sweeps/run_blur_sweep.py -- generates blur_sweep.csv; source of P0, n_prime, vacuum wavelength, and the ASM simulation grid used throughout Sections 2.2-5.
- stage2_fixed/myopia_axial.py, stage2_fixed/phase2_retina.py -- reduced-eye f'=n'/P0 convention and the media-reduced-wavelength justification used in Section 2.3.
- ExperimentalComparison/analyze.py, run_pipeline.py -- ExP2 DNG measurement pipeline (Section 3).
- ExperimentalComparison/rss_comparison.py -- RSS model implementation and RSS-vs-ASM comparison (Section 4).
- ExperimentalComparison/clinical_and_two_pinhole_check.py -- clinical anchor check and two-pinhole target computation (Section 5).
- Array_Design/array_geometry_lab/array_calculation/engine/ghost_formula.py -- validated ghost-separation formula (R^2=0.995135), reused verbatim in Section 2.4/5.2.
- Array_Design/array_geometry_lab/array_evaluation/comparison_metric/gates.py -- margin and safety-threshold framework, reused in Section 2.4/5.2.
- ExperimentalComparison/ExP3_frosted_screen/analyze_exp3.py, build_exp3_report.py -- ExP3 frosted-screen measurement pipeline and report (Section 6).

Underlying data tables (all under ExperimentalComparison/)

- measurements/all_measurements.csv -- ExP2 per-file measured/predicted comparison (Section 3).
- measurements/rss_vs_asm_comparison.csv -- RSS-vs-ASM per-row comparison (Section 4).
- measurements/nopinhole_control.csv -- ExP2 no-pinhole control series (Section 3.3).
- report/two_pinhole_experiment_targets.csv -- two-pinhole shot-list table (Section 5.2).
- ExP3_frosted_screen/measurements/all_spot_measurements.csv, excluded_files.csv -- ExP3 raw measurement and exclusion tables (Section 6).
- new_sweeps/blur_sweep.csv -- the single ASM theory table used by every ASM comparison in this report.

This document was generated directly from the CSV/table outputs listed above; every numeric value shown in Sections 3-6 is recomputed from those files at render time by ExperimentalComparison/final_report/build_final_report.py, not retyped by hand.